

Diffusion tensor imaging with P2ROUD

From EPI k-space to color-coded fiber-direction maps

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1. Overview

P2ROUD reconstructs diffusion-weighted EPI scans from MR Solutions raw data, corrects the typical EPI artefacts and exports the images for tensor analysis and tractography.

Example: mouse brain DTI, study 2482, scan 32518 (34 slices, 130 volumes, $b = 0, 1000$ and 2000 s/mm²).

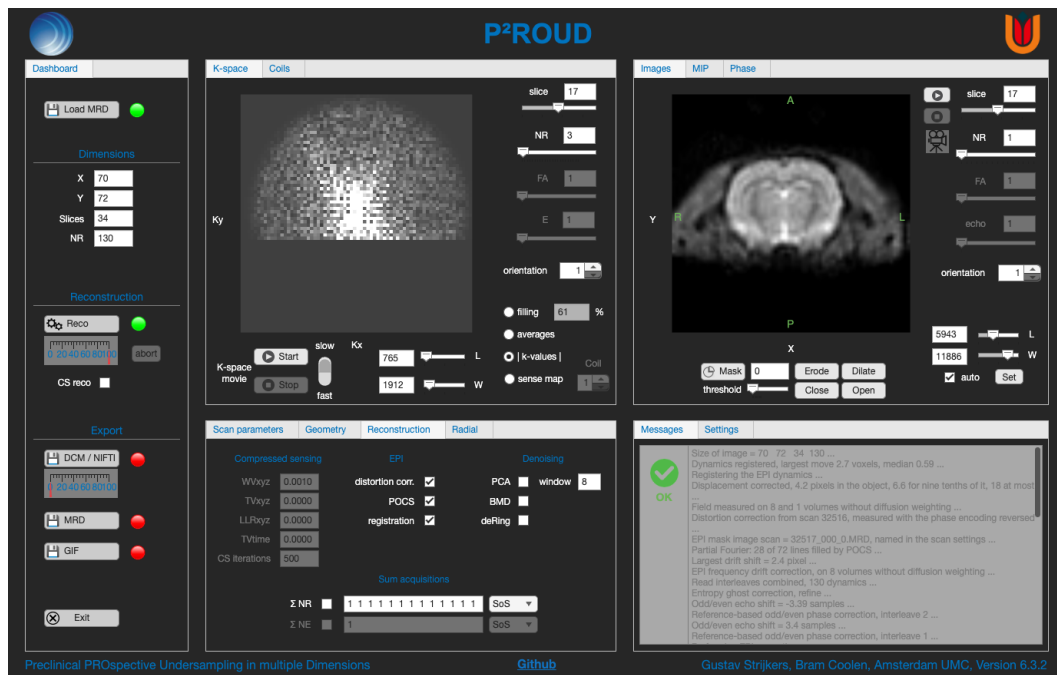


Figure 1. P2ROUD with the reconstructed DTI scan. The EPI corrections are options in the Reconstruction tab.

2. Pipeline

Reco runs these steps automatically:

1. **Ghost correction** – removes the N/2 ghost.
2. **Drift correction** – removes image shifts caused by frequency drift.
3. **Partial Fourier (POCS)** – fills in the k-space lines that were not acquired.
4. **Distortion correction** – like FSL topup, using a $b = 0$ scan with reversed phase encoding.
5. **Registration** – aligns all volumes.

Export writes DICOM and NIFTI files, plus scripts for MRtrix3 and DSI Studio.

3. Effect of the corrections

Distortion correction. EPI images are compressed or stretched along the phase-encoding direction. After correction the EPI matches the undistorted FLASH image (Figure 2).

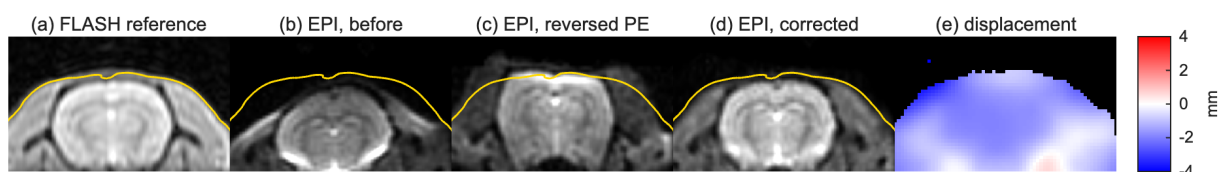


Figure 2. (a) FLASH reference, (b) EPI before correction, (c) EPI with reversed phase encoding, (d) EPI after correction, (e) estimated displacement. Yellow: head outline from (a).

Registration. The diffusion-weighted volumes shift, depending on the gradient direction (eddy currents). Registration removes these shifts and sharpens the result (Figure 3).

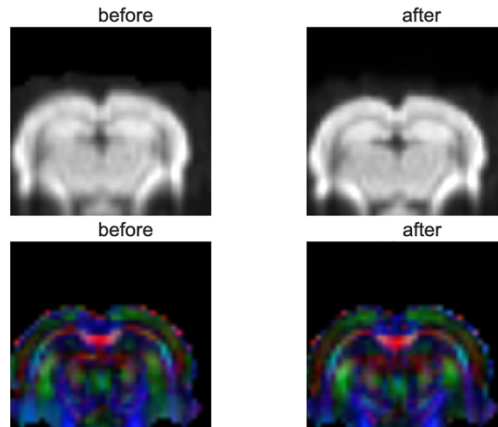


Figure 3. Mean b = 2000 image and color map, before and after registration.

4. Result

Color-coded maps of the principal diffusion direction, weighted by fractional anisotropy (Figure 4).

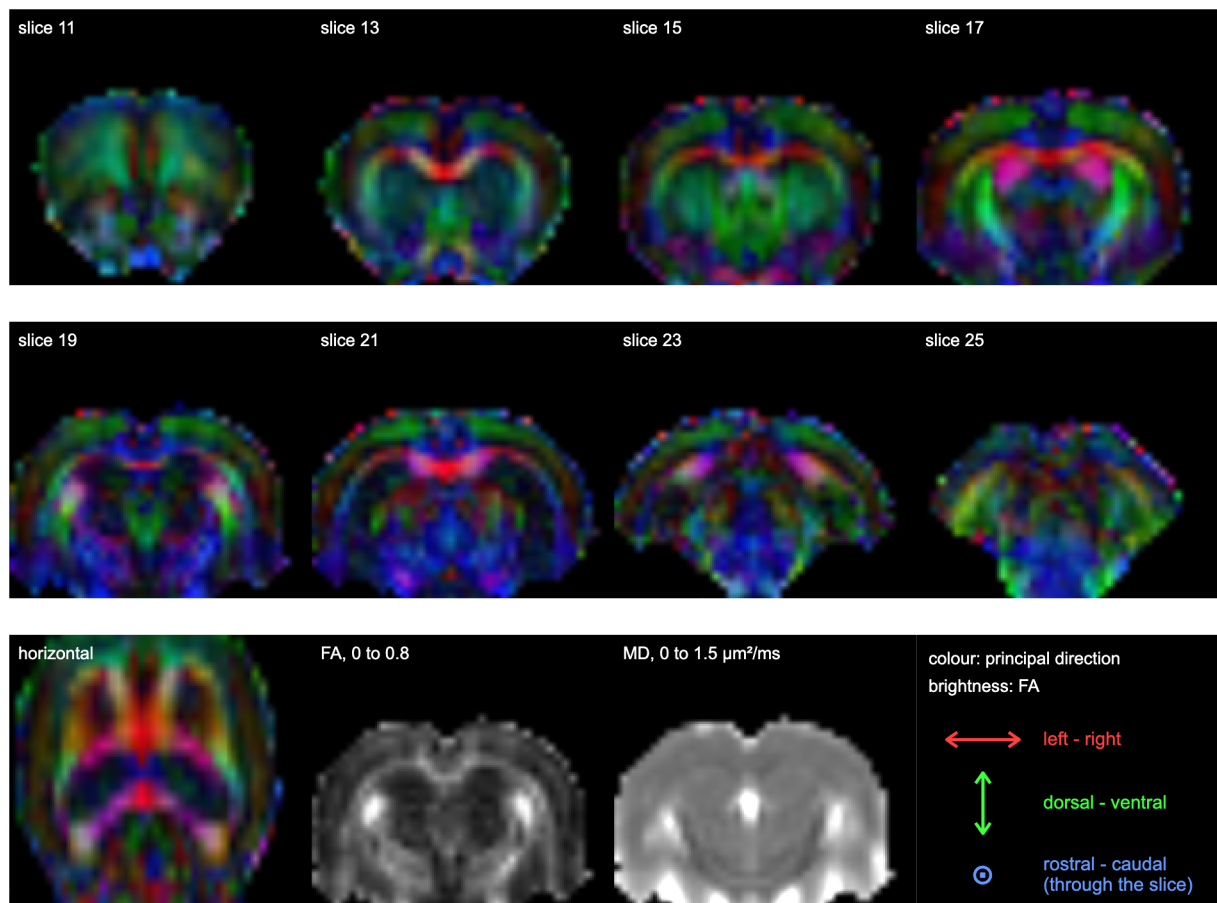


Figure 4. Principal-eigenvector (v1) maps: coronal slices from rostral to caudal, a horizontal section, FA and MD maps, and the color key.